

# HireAgent

## AI Recruitment Automation Platform

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### System Design Document

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## 1. Executive Summary

HireAgent is an end-to-end AI recruitment automation platform that transforms the hiring process from job posting to offer letter. Built on top of TalentPing's proven cold outreach and email deliverability infrastructure, HireAgent extends the pipeline to cover the full recruitment lifecycle: resume screening, candidate sourcing, interview scheduling, assessment management, and hiring pipeline orchestration.

The recruitment industry faces a fundamental efficiency problem. Recruiters spend 70% of their time on repetitive tasks -- screening resumes, scheduling interviews, and sending follow-up emails -- leaving only 30% for high-value activities like candidate evaluation and relationship building. HireAgent automates the repetitive 70% while augmenting the evaluative 30%.

The platform serves three user segments: in-house HR teams at growing companies (50-500 employees), recruitment agencies managing multiple client mandates, and staffing firms handling high-volume hiring. Each segment benefits from HireAgent's core AI capabilities but with workflows tailored to their specific operational needs.

**Competitive Advantage:** Unlike standalone ATS tools, HireAgent combines sourcing, outreach, screening, and pipeline management in a single platform. The TalentPing email infrastructure provides industry-leading deliverability for candidate outreach, while the AI screening engine reduces time-to-shortlist by 80%.

**Market Opportunity:** The global recruitment software market is projected to reach \$3.2 billion by 2028. HireAgent targets the underserved SMB segment where budgets preclude enterprise ATS tools but hiring volume demands automation.

## 2. System Architecture

### 2.1 Technology Stack

| Layer                | Technology                | Purpose                                             |
|----------------------|---------------------------|-----------------------------------------------------|
| Runtime              | Python 3.11+              | Backend services, AI pipeline                       |
| Backend Framework    | FastAPI                   | REST API, WebSocket, background jobs                |
| Frontend             | React / Next.js 14        | Recruiter dashboard, candidate portal               |
| Primary Database     | PostgreSQL 16             | Job, candidate, and pipeline data                   |
| Cache / Queue        | Redis 7 + BullMQ          | Outreach queue, session cache, real-time events     |
| Vector Store         | Qdrant                    | Resume embeddings, job description embeddings       |
| AI Gateway           | OpenRouter                | LLM for screening, matching, and content generation |
| Email Infrastructure | TalentPing Engine         | Warm-up, deliverability, multi-sender rotation      |
| Calendar Integration | Google Calendar + Outlook | Interview scheduling and availability               |
| File Processing      | Apache Tika + Custom      | Resume parsing (PDF, DOCX, images)                  |

### 2.2 AI Pipeline Architecture

The AI pipeline processes candidates through a multi-stage evaluation funnel:

#### Resume Processing Pipeline

1. Resume Ingestion: Candidate uploads resume (PDF/DOCX) or it is sourced from job boards. The file is stored in S3-compatible storage.
2. Text Extraction: Apache Tika extracts text from the resume, handling multi-column layouts, tables, and embedded images via OCR.
3. Structured Parsing: LLM extracts structured data -- contact info, work experience (company, role, duration), education, skills, certifications, and projects.
4. Skill Extraction: A fine-tuned classifier maps extracted skills to a standardized skill taxonomy (e.g., 'React.js' and 'ReactJS' both map to 'React').
5. Embedding Generation: The parsed resume is embedded into a vector representation and stored in Qdrant for semantic matching.
6. Candidate Scoring: For each open job, candidates are scored on skill match (40%), experience relevance (30%), education fit (15%), and cultural indicators (15%).

#### Outreach Engine (TalentPing Integration)

HireAgent reuses TalentPing's battle-tested email infrastructure for candidate outreach. This includes multi-domain sender rotation, automated warm-up sequences, bounce and reply detection, and deliverability optimization. The integration provides 95%+ inbox placement rates for candidate outreach emails.

## 2.3 Integration Layer

HireAgent integrates with major job boards and HR tools through a unified adapter pattern:

| Integration       | Type                 | Direction     | Purpose                                |
|-------------------|----------------------|---------------|----------------------------------------|
| Naukri.com        | Job Board            | Bidirectional | Post jobs, import applications         |
| LinkedIn          | Professional Network | Import        | Candidate sourcing, profile enrichment |
| Indeed            | Job Board            | Bidirectional | Post jobs, import applications         |
| Google Calendar   | Calendar             | Bidirectional | Interview scheduling, availability     |
| Outlook Calendar  | Calendar             | Bidirectional | Interview scheduling, availability     |
| WhatsApp Business | Messaging            | Outbound      | Candidate communication, reminders     |
| Slack             | Collaboration        | Outbound      | Hiring team notifications              |
| DocuSign / Digio  | E-signature          | Outbound      | Offer letter signing                   |

## 3. Data Model

### 3.1 Core Tables

| Table              | Description                   | Key Fields                                                                                    |
|--------------------|-------------------------------|-----------------------------------------------------------------------------------------------|
| organizations      | Hiring companies and agencies | id, name, type (company/agency), industry, size, plan, settings_json                          |
| jobs               | Open positions                | id, org_id, title, department, location, type (full-time/contract), requirements_json, status |
| candidates         | Candidate profiles            | id, name, email, phone, current_company, current_role, experience_years, skills_json          |
| applications       | Job-candidate associations    | id, job_id, candidate_id, source, stage, score, applied_at, status                            |
| resumes            | Parsed resume data            | id, candidate_id, file_path, parsed_json, skills_extracted, embedding_id                      |
| screenings         | AI screening results          | id, application_id, skill_match, experience_score, overall_score, reasoning                   |
| interviews         | Scheduled interviews          | id, application_id, type (phone/video/onsite), scheduled_at, interviewer_ids, status          |
| outreach_sequences | Email campaign sequences      | id, job_id, template_ids, sender_id, status, stats_json                                       |
| email_accounts     | Sender email accounts         | id, org_id, email, provider, warmup_status, daily_limit, reputation_score                     |
| assessments        | Skill assessment results      | id, application_id, type, score, responses_json, completed_at                                 |
| offer_letters      | Generated offers              | id, application_id, template_id, salary, benefits_json, status, signed_at                     |
| analytics          | Recruitment metrics           | id, org_id, job_id, date, applications, screenings, interviews, offers, hires                 |

All tables include org\_id for multi-tenant isolation with PostgreSQL RLS. Candidate PII is encrypted at the column level. Soft delete via deleted\_at is enforced across all tables. The resumes table stores both the raw file reference and the structured parsed output for quick access.

## 4. Core Features

### 4.1 AI Resume Parsing and Ranking

The resume parsing engine processes PDF, DOCX, and image-based resumes with 95%+ accuracy for structured data extraction. The ranking algorithm evaluates candidates against job requirements using a weighted scoring model. Recruiters can customize weights for different criteria and set minimum thresholds for automatic advancement.

The system handles common challenges: career gaps (contextual analysis rather than penalization), non-traditional backgrounds (skill-based matching over credential matching), and international experience (currency and title normalization).

### 4.2 Multi-Channel Outreach

Candidate outreach spans three channels, orchestrated through configurable sequences:

1. Email: Personalized multi-step sequences using TalentPing infrastructure. A/B testing of subject lines and content. Automatic reply detection and sequence pausing.
2. WhatsApp: Template-based messages for interview confirmations, document requests, and status updates. Two-way communication with recruiter routing for complex queries.
3. LinkedIn: InMail-style outreach for passive candidates (via LinkedIn Recruiter integration). Connection request automation with personalized notes.

### 4.3 Automated Interview Scheduling

HireAgent eliminates the back-and-forth of interview scheduling. The system reads interviewer availability from Google Calendar or Outlook, presents available slots to the candidate via email or WhatsApp, and books the interview once the candidate selects a slot. Automatic reminders are sent 24 hours and 1 hour before the interview. Rescheduling and cancellation are self-service for candidates.

### 4.4 Video Screening with AI Analysis

Asynchronous video screening allows candidates to record responses to pre-set questions at their convenience. The AI analyzes responses for communication clarity (speech-to-text analysis), technical accuracy (keyword and concept matching against expected answers), confidence indicators, and overall presentation. Results are presented as a scorecard for the recruiter's review.

### 4.5 Candidate Pipeline Dashboard

A Kanban-style pipeline board showing candidates flowing through stages: Sourced, Applied, Screened, Interviewed, Assessed, Offered, Hired. Drag-and-drop to move candidates between stages. Bulk actions for advancing, rejecting, or messaging multiple candidates. Real-time updates via WebSocket when team members make changes.

### 4.6 Job Board Aggregation and Posting

Post job openings to multiple job boards (Naukri, Indeed, LinkedIn) from a single interface. Incoming applications from all boards are aggregated into the HireAgent pipeline with source tracking. Job

description optimization suggestions powered by AI to improve application rates and reduce biased language.

#### **4.7 Offer Letter Generation with E-Signature**

Generate professional offer letters from customizable templates with auto-populated candidate and compensation details. Integration with DocuSign and Digio (for India) enables digital signature collection. Offer tracking shows when the letter was sent, viewed, and signed.

#### **4.8 Onboarding Checklist Automation**

Once a candidate accepts the offer, HireAgent generates a pre-boarding checklist: document collection (ID proof, education certificates, previous employer relieving letter), IT provisioning requests, welcome email scheduling, and Day 1 agenda. Progress is tracked and reminders are sent automatically.

#### **4.9 Diversity and Bias Detection**

AI-powered analysis of job descriptions to detect potentially biased language (gendered terms, unnecessary requirements that disproportionately exclude underrepresented groups). Pipeline analytics show diversity metrics at each stage to identify where drop-off occurs. All features are opt-in and designed to assist, not automate, DEI decisions.

#### **4.10 Referral Tracking**

Employees can submit referrals through a dedicated portal. Referred candidates are tagged in the pipeline with referrer attribution. Referral bonus tracking with automatic payout triggers when the referred candidate completes the probation period.

## 5. AI Models

| Model                 | Purpose                              | Input                                | Output                                   | Performance Target                     |
|-----------------------|--------------------------------------|--------------------------------------|------------------------------------------|----------------------------------------|
| Resume Parser         | Extract structured data from resumes | Raw resume text/image                | JSON with experience, skills, education  | >95% field accuracy                    |
| Skill Extractor       | Map skills to taxonomy               | Parsed resume text                   | Standardized skill tags with proficiency | >90% mapping accuracy                  |
| Job-Candidate Matcher | Score candidate fit for job          | Job requirements + candidate profile | Match score (0-100) with reasoning       | >85% agreement with recruiter ranking  |
| Interview Analyzer    | Evaluate video responses             | Transcribed interview audio          | Scorecard with dimension ratings         | >80% correlation with human evaluation |
| JD Optimizer          | Improve job descriptions             | Draft job description                | Optimized JD with bias flags             | 15%+ improvement in application rate   |

All models run through OpenRouter using free-tier models for cost efficiency. Model outputs include confidence scores and reasoning chains for transparency. Low-confidence results (< 70%) are flagged for human review. Models are fine-tuned on anonymized recruitment data contributed by partner organizations.

## 6. API Design

### 6.1 RESTful Endpoints

| Method | Endpoint                             | Description                                    |
|--------|--------------------------------------|------------------------------------------------|
| POST   | /api/v1/jobs                         | Create a new job posting                       |
| GET    | /api/v1/jobs/:id/candidates          | List candidates for a job with scores          |
| POST   | /api/v1/candidates                   | Add a candidate (manual or parsed from resume) |
| POST   | /api/v1/candidates/bulk-upload       | Bulk resume upload and parsing                 |
| PUT    | /api/v1/applications/:id/stage       | Move candidate to a pipeline stage             |
| POST   | /api/v1/interviews/schedule          | Schedule an interview                          |
| GET    | /api/v1/interviews/:id/analysis      | Get AI analysis of interview recording         |
| POST   | /api/v1/outreach/sequences           | Create an outreach sequence                    |
| GET    | /api/v1/outreach/sequences/:id/stats | Get outreach sequence performance              |
| POST   | /api/v1/offers/generate              | Generate an offer letter                       |
| GET    | /api/v1/analytics/pipeline           | Pipeline conversion analytics                  |
| GET    | /api/v1/analytics/diversity          | Diversity metrics across pipeline              |

All endpoints require JWT authentication with organization context. Webhook endpoints are provided for integration partners (job boards, calendar services). Rate limiting is enforced at 200 requests/minute per organization.



## 7. Revenue Model

### 7.1 Pricing Tiers

| Plan       | Monthly Fee | Per-Hire Fee | Active Jobs | Features                                           |
|------------|-------------|--------------|-------------|----------------------------------------------------|
| Startup    | \$199/mo    | \$99/hire    | 5           | Resume parsing, pipeline board, basic outreach     |
| Growth     | \$499/mo    | \$199/hire   | 20          | AI screening, video interviews, calendar sync      |
| Business   | \$999/mo    | \$299/hire   | Unlimited   | Full outreach suite, assessments, analytics        |
| Enterprise | Custom      | \$499/hire   | Unlimited   | Dedicated infrastructure, API, custom integrations |

AI screening credits are included in each plan (100/500/2000/unlimited respectively). Additional credits can be purchased at \$0.50 per resume screening. Annual contracts receive a 20% discount. Recruitment agencies get special volume pricing with per-client billing.

## 8. Security

### 8.1 Candidate Data Protection

Resume and candidate PII is encrypted at rest using AES-256 column-level encryption. Data retention policies are configurable per organization (default: 24 months from last activity). Candidates can request data access and deletion through a self-service portal (GDPR/DPDP compliance).

### 8.2 Consent Management

Candidate consent is tracked at every stage: resume processing consent, communication consent (per channel), data sharing consent (between agency and client), and assessment consent. Consent records are immutable and auditable.

### 8.3 Access Control

Role-based access with four default roles: Admin (full access), Hiring Manager (job-specific access), Recruiter (candidate management), and Interviewer (view assigned interviews only). Organization admins can create custom roles with granular permissions.

## 9. Testing Strategy

### 9.1 Test Categories

| Category          | Scope                      | Tools                   | Coverage Target    |
|-------------------|----------------------------|-------------------------|--------------------|
| Unit Tests        | Parsers, scorers, matchers | pytest, unittest.mock   | >85% line coverage |
| Integration Tests | API, database, queue       | pytest + testcontainers | >80% path coverage |

|                      |                                     |                                  |                                   |
|----------------------|-------------------------------------|----------------------------------|-----------------------------------|
|                      | operations                          |                                  |                                   |
| Resume Parsing Tests | Accuracy across formats and layouts | Custom test harness + golden set | >95% on 500-resume test set       |
| Matching Tests       | Candidate-job matching accuracy     | Custom eval suite                | >85% agreement with human ranking |
| Deliverability Tests | Email inbox placement               | GlockApps, mail-tester           | >95% inbox rate                   |
| E2E Tests            | Full hiring workflow                | Playwright                       | All critical workflows            |

## 9.2 Resume Parsing Golden Set

A curated set of 500 resumes across various formats (PDF, DOCX, image), layouts (single-column, multi-column, creative), and career backgrounds (tech, non-tech, fresh graduate, executive) serves as the golden test set. Each resume has manually verified structured output. New parser versions must achieve  $\geq 95\%$  field-level accuracy on this set before deployment.

## 10. Milestones

| Phase  | Timeline    | Deliverables                                                 | Success Criteria                      |
|--------|-------------|--------------------------------------------------------------|---------------------------------------|
| MVP    | Weeks 1-5   | Resume parsing, candidate pipeline, basic outreach (email)   | 100 resumes parsed with >90% accuracy |
| Alpha  | Weeks 6-7   | AI screening, job board integration (Naukri), scheduling     | 10 pilot companies, 50+ jobs posted   |
| Beta   | Weeks 8-9   | Video screening, multi-channel outreach, offer generation    | 500 candidates processed, 20+ hires   |
| Polish | Weeks 10-11 | Analytics, diversity tools, onboarding checklists, referrals | NPS > 40 from pilot users             |
| GA     | Weeks 12-13 | Production hardening, billing, marketing site, documentation | Public launch with SLA                |

The MVP prioritizes resume parsing accuracy and pipeline management as these are the highest-value features for early adopters. TalentPing's outreach infrastructure is integrated from Week 1, providing immediate email deliverability without cold start.

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